



Improvements to the post-processing of weather forecasts using machine learning and feature selection

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ABSTRACT

This study aims to develop and improve machine learning-based post-processing models for precipitation, temperature, and wind speed predictions using the Mesoscale Model (MSM) dataset provided by the Japan Meteorological Agency (JMA) for 18 locations across Japan, including plains, mountainous regions, and islands. By incorporating meteorological variables from grid points surrounding the target locations as input features and applying feature selection based on correlation analysis, we found that, in our experimental setting, the LightGBM-based models achieved lower RMSE than the specific neural-network baselines tested in this study, including a reproduced CNN baseline, and also generally achieved lower RMSE than both the raw MSM forecasts and the JMA post-processing product, MSM Guidance (MSMG), across many locations and forecast lead times. Because precipitation has a highly skewed distribution with many zero cases, we additionally examined Tweedie-based loss functions and event-weighted training strategies for precipitation forecasting. These improved event-oriented performance relative to the original LightGBM model, especially at higher rainfall thresholds, although the gains were site dependent and overall performance remained slightly below MSMG.

1. Introduction

1.1. Research background and objectives

Numerical weather prediction (NWP) is a primary method of weather forecasting that represents future atmospheric states using mathematical models based on observation data. Over the years, it has evolved through advancements in the representation of physical processes, model initialization, including assimilation, and the introduction of ensemble modeling (Bauer et al., 2015). However, NWP is not without its limitations, such as forecast errors and biases. While forecast variance might be unavoidable, several post-processing techniques using machine learning models have been developed to mitigate these biases (Liu et al., 2023; Rojas-Campos et al., 2023; Yoshikane and Yoshimura, 2022; Zhang and Ye, 2021; Kudo, 2022; Peng et al., 2020; Tang et al., 2021; Salazar et al., 2022; Xu et al., 2020).

The Japan Meteorological Agency (JMA) has also introduced a post-processing method called the MSM Guidance (MSMG) to correct the errors of the Mesoscale Model (MSM), which has a finer grid spacing (5 km) and targets areas around Japan compared to global models (JMA, 2024). In MSMG, JMA primarily employs statistical methods, such as the Kalman filter, frequency bias correction, and neural networks, to reduce the systematic biases in MSM.

This study aims to develop machine learning-based post-processing models for precipitation, temperature, and wind speed predictions – representative meteorological variables – using MSM data for 18 locations across Japan, including plains, mountainous regions, and islands. Meteorological variables from grid points surrounding the prediction locations were used as input features, and feature selection based on correlation analysis was applied. In our experimental setting, the LightGBM-based models achieved lower RMSE than the tested neural-network baselines, including the reproduced CNN baseline, and also generally achieved lower RMSE than both the raw MSM forecasts and the JMA post-processing product, MSM Guidance (MSMG). Among the LightGBM-based models, those using surrounding-grid information with correlation-based feature selection showed the lowest RMSE across many locations and forecast lead times.

For precipitation, we further examined Tweedie-based loss functions and event-weighted training strategies, which improved event-oriented metrics for some sites and rainfall thresholds even when the gains in RMSE were limited.

1.2. Related work

Several studies have developed post-processing models for precipitation using methods such as convolutional neural networks (CNNs),

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neural networks, and Support Vector Machines (Zhang and Ye, 2021; Liu et al., 2023; Rojas-Campos et al., 2023; Yoshikane and Yoshimura, 2022). Among them, Zhang and Ye (2021) conducted comparative experiments on machine learning models, input parameters, and training data periods for precipitation prediction, and concluded that LightGBM provides the most balanced performance among the tested models.

For temperature post-processing models, CNNs, neural networks, and LightGBM have been employed (Kudo, 2022; Peng et al., 2020; Tang et al., 2021). Kudo (2022) developed a CNN-based temperature post-processing model targeting the Kanto region of Japan and demonstrated its superior performance over the JMA's MSMG.

Wind speed post-processing models have also been explored using neural networks and LightGBM (Salazar et al., 2022; Tang et al., 2021; Xu et al., 2020). Xu et al. (2020) utilized LightGBM for wind speed prediction, analyzing feature importance derived from the model. Their study revealed that using all features, including various weather elements, as input outperformed models where features were limited primarily to wind speed.

In addition to using NWP outputs, some studies have leveraged meteorological forecast data from existing weather forecasting services (Iwase and Takenawa, 2024; Tsipis et al., 2023). Iwase and Takenawa (2024) employed machine learning models such as LightGBM, XGBoost, and neural networks to predict temperature and precipitation in mountainous regions. By incorporating surrounding weather forecast data as input variables, the results were superior to existing weather forecast services.

In recent machine learning research, deep learning methods have been developed significantly in fields such as natural language processing and image recognition. In contrast, for tabular data, tree-based models currently outperform deep learning models in terms of accuracy and require less hyperparameter tuning costs (Shwartz-Ziv and Armon, 2022; Grinsztajn et al., 2022). The utility of tree-based models has also been demonstrated in post-processing NWP outputs, which often involve tabular data (Zhang and Ye, 2021; Xu et al., 2020). Recently, Hieta and Partio (2025) reported that a tree-based gradient-boosting post-processing scheme (XGBoost) reduces RMSE for short-range near-surface forecasts.

The remainder of this paper is organized as follows. Section 2 defines the methods and data used in this study. Section 3 presents the results and discussion. Finally, Section 4 concludes the study.

2. Methods

2.1. Data

In this study, we use MSM data from the JMA as the input for the model and observation data as the target data for training. The MSM data were collected and distributed by Research Institute for Sustainable Humanosphere, Kyoto University (RISH, 2024). The observation data from JMA include cumulative precipitation over the past three hours, hourly temperature, and wind speed. We selected the 18 observation sites from all over Japan (Fig. 1, Table 1). Additionally, in order to compare the forecast results, we collected MSMG data, a post-processed forecast provided by JMA and distributed by the Japan Meteorological Business Support Center (JMBSC).

MSM consists of three-dimensional gridded predictions of future atmospheric conditions, including temperature, wind, moisture, and solar radiation, with fine grid spacing of approximately 5 km. The forecasts cover Japan and surrounding sea areas (JMA, 2024). Surface-level data are available at hourly intervals, while pressure-level data are available at three-hour intervals. The forecasts are issued every three hours (00, 03, 06, 09, 12, 15, 18, 21 UTC) and extend up to 39 h ahead. The surface grid comprises 505×481 points (grid spacing: 0.05° latitude $\times$ 0.0625° longitude), while the pressure-level grid consists of 253×241 points (grid spacing: 0.1° latitude $\times$ 0.125° longitude),

Table 1

Latitude, longitude, altitude, and station numbers of the 18 sites shown in Fig. 1.

	Latitude	Longitude	Altitude	Station number
Ashimine	26.193	127.638	3	0
Uchinoura	31.277	131.055	8	1
Minamiaso	32.832	131.013	394	2
Uwanokogen	35.432	134.583	540	3
Imabari	34.053	132.975	27	4
Sakai	34.555	135.485	20	5
Kuwana	35.050	136.693	3	6
Sekigahara	35.363	136.467	130	7
Itoigawa	37.043	137.875	8	8
Nobeyama	35.948	138.472	1350	9
Niijima	34.368	139.268	29	10
Tsujido	35.320	139.450	5	11
Saitama	35.875	139.587	8	12
Kusatsu	36.617	138.592	1223	13
Kamaishi	39.270	141.878	5	14
Oma	41.527	140.912	14	15
Ishikari	43.193	141.370	5	16
Kamikawa	43.847	142.753	324	17

Table 2

Meteorological elements used as predictors, including names and units. A ✓ indicates that the element is used at the surface and/or at the representative pressure levels (850, 500, and 200 hPa). (Relative humidity is used at 850/500 hPa only.)

Element name	Surface (hourly)	Pressure level (every 3 h)	Unit
Temperature	✓	✓	°C
Relative humidity	✓	850/500 hPa only	%
Hourly precipitation	✓		mm
Wind speed	✓	✓	m s ⁻¹
U wind component	✓	✓	m s ⁻¹
V wind component	✓	✓	m s ⁻¹
Vertical velocity		✓	hPa s ⁻¹
Surface pressure	✓		hPa
Sea level pressure	✓		hPa
Geopotential height		✓	m
Solar radiation	✓		W m ⁻²
Total cloud cover	✓		%
Low cloud cover	✓		%
Mid-level cloud cover	✓		%
High cloud cover	✓		%

covering the area from the northwest corner (47.6° N, 120° E) to the southeast corner (22.4° N, 150° E).

MSM outputs a variety of meteorological elements on the surface and on multiple standard pressure levels. In this study, we use the elements listed in Table 2 as predictors. For pressure-level variables, although MSM provides outputs at more pressure levels, we use only three representative levels (850, 500, and 200 hPa), following prior work (Liu et al., 2023), to capture information from the lower, middle, and upper atmosphere while reducing dimensionality. The CNN baseline uses a different input configuration following prior work (including additional near-surface pressure levels); see Section 2.5.

For each station, we train an independent regression model. Except for the CNN-based model, the inputs to LightGBM and the neural network are one-dimensional feature vectors constructed by concatenating (i) surface variables within a temporal window ($t, t-1, t-2$), (ii) pressure-level variables at the target time t only (850, 500, and 200 hPa), and (iii) auxiliary scalar features (lead time, month, and hour of day). All gridded predictors are flattened before concatenation. The temporal window length was chosen to match the 3-hourly forecast cycle of MSM/MSMG, so that a consistent set of predictors is available across lead times and forecast cycles.

The input variables consist of 13 surface variables and 7 pressure-level variables as summarized in Table 2. Among the pressure-level variables, relative humidity is available only at two levels (850 and

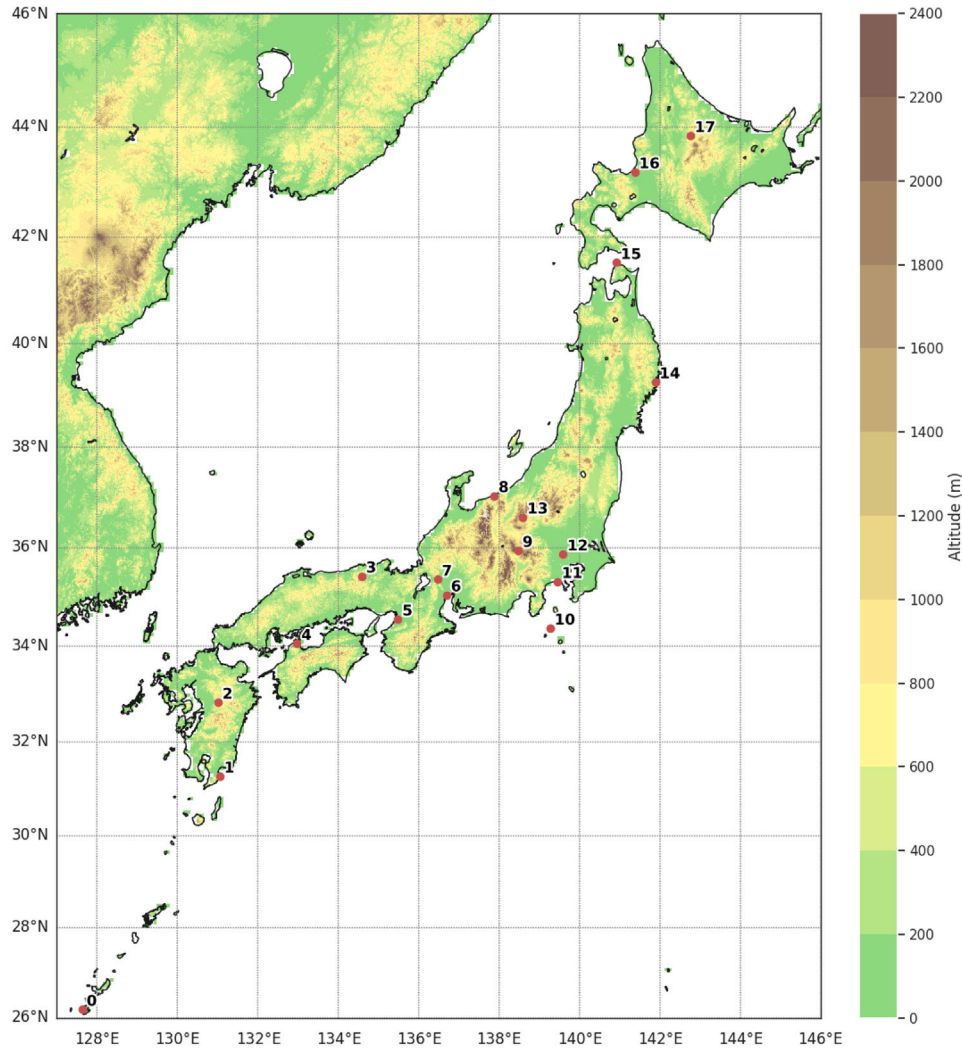


Fig. 1. Map showing the 18 selected locations.

500 hPa), whereas the other six variables are available at all three levels.

We prepared two input configurations. In the first, only the grid cell closest to the prediction point is used. In this configuration, the total number of input features is 62, computed as

$$13 \times 3 + (6 \times 3 + 2) + 3 = 62,$$

where 13×3 accounts for the 13 surface variables at three time steps, $(6 \times 3 + 2)$ counts pressure-level combinations (six variables at three levels plus relative humidity at two levels) at time t , and $+3$ accounts for lead time, month, and hour of day.

In the second, surrounding grid cells are included to assess the benefit of spatial context. Because the pressure-level fields have twice the horizontal grid spacing of the surface fields, we use an 11×11 grid for surface variables and a 7×7 grid for pressure-level variables so that the covered areas are comparable. Under this surrounding-grid configuration, the total number of input features is 5702, computed as

$$13 \times 3 \times 11 \times 11 + (6 \times 3 + 2) \times 7 \times 7 + 3 = 5702,$$

where 13 is the number of surface variables, 3 is the number of time steps for surface variables, 11×11 is the surface grid, $(6 \times 3 + 2)$ counts pressure-level combinations at time t , 7×7 is the pressure-level grid, and $+3$ accounts for lead time, month, and hour of day.

The output variables are the cumulative precipitation over the past three hours, as well as hourly temperature and wind speed. Observation

data from 2019 to 2021 were used for training, 2022 for validation, and 2023 for testing, as shown in Table 3. Since the MSM provides complete fields without missing values, only observation samples containing missing values were excluded from the dataset. The validation data were used solely for model selection and hyperparameter tuning, and were not used to retrain the model prior to testing. Fig. 2 presents box plots for each location and meteorological variable.

For MSMG, the accumulated surface precipitation over the past three hours and hourly surface temperature and wind speed are predicted up to 39 h ahead, with forecast outputs available every three hours (00, 03, 06, 09, 12, 15, 18, and 21 UTC). Precipitation forecasts are provided on a grid with dimensions of 560 (longitude) $\times$ 480 (latitude), corresponding to a grid spacing of 0.05° in latitude and 0.0625° in longitude, covering the area from $(47.975^\circ\text{N}, 120.03125^\circ\text{E})$ in the northwest to $(20.025^\circ\text{N}, 149.96875^\circ\text{E})$ in the southeast. In contrast, forecasts for temperature and wind speed are provided at the locations of JMA observation stations.

The dataset is summarized as follows. The outputs consist of three variables (precipitation, temperature, and wind speed), while the inputs are identical across them. Missing values occur only in the observation data. The period covers the entire span of the training (2019–2021), validation (2022), and test (2023) sets. The total number of rows is determined by the combination of period, forecast cycles, lead times, and locations, while the number of columns differs according to the input configuration.

Table 3

The number of samples in the training, validation, and test datasets, and the total number of features when using the surrounding grids of the prediction point, as well as the number of features after feature selection based on correlation analysis.

	Train samples	Validation samples	Test samples	Original features	Dropped features
Ashimine	113 854	37 882	37 934	5702	239
Uchinoura	113 854	37 843	37 947	5702	296
Minamiaso	113 828	37 830	37 960	5702	502
Uwanokogen	108 642	37 570	37 947	5702	385
Imabari	113 802	37 830	37 934	5702	408
Sakai	113 776	37 830	37 947	5702	455
Kuwana	111 891	37 817	37 960	5702	354
Sekigahara	113 568	37 830	37 934	5702	437
Itoigawa	113 609	37 843	37 817	5702	516
Nobeyama	113 022	37 544	37 791	5702	869
Nijijima	113 477	37 778	37 921	5702	236
Tsujido	113 763	37 791	37 960	5702	416
Saitama	113 789	37 817	37 960	5702	248
Kusatsu	112 359	37 271	37 752	5702	733
Kamaishi	113 464	37 687	37 921	5702	295
Oma	113 815	37 804	37 895	5702	256
Ishikari	113 243	37 791	37 934	5702	271
Kamikawa	113 607	36 907	37 791	5702	413

Period: 2019–2023 (1826 days)

Forecast cycles per day: 8 (3-hourly)

Lead times per cycle: 13 (3–39 h)

Locations: 18

Total rows: 3,418,272 (before removing missing observations)

2.2. LightGBM

Several previous studies on tabular data and post-processing model development have reported that decision tree-based models outperform other machine learning models in terms of training cost and performance scores. Among decision tree-based models, LightGBM (Ke et al., 2017) is particularly fast and high-performing (Iwase and Takenawa, 2024). Therefore, this study adopts LightGBM for the development of the post-processing model. LightGBM is an ensemble model based on decision trees, employing a boosting algorithm that sequentially adds decision trees and uses a quadratic optimization method.

Since LightGBM cannot handle multiple output variables simultaneously, a separate model should be created for each output variable. During training, Early Stopping was applied using validation data with a patience of 10 epochs, and the loss function was set to Mean Squared Error (MSE).

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 \quad (1)$$

where N is the number of data points, $\hat{y}_i$ is the predicted value, and y_i is the observed value.

2.3. Feature selection

When surrounding grids are included, the number of features reached 5702, which is excessive for LightGBM. Feature selection is necessary to reduce training time and prevent overfitting. To achieve this, the correlation analysis method proposed in Bedi and Toshniwal (2018), denoted as FS0 hereafter, was applied. FS0 computes pairwise absolute Pearson correlations among candidate features and removes redundancies.

To avoid removing critical input information, we exclude features at the grid closest to the prediction target and calendar variables from the analysis by dropping columns. Thus, the correlation analysis is performed solely among features from surrounding grids.

Let τ denote the correlation threshold. FS0 builds the absolute correlation matrix $[|r(x_i, x_j)|]_{i,j}$ and scans features in their column order.

For each feature x_j , if there exists any earlier feature x_i ($i < j$) with $|r(x_i, x_j)| > \tau$, then x_j is dropped (left-to-right rule). Based on validation data, the threshold was set to $\tau = 0.9$. The number of features after correlation analysis for each location is summarized in Table 3.

After applying the aforementioned feature selection process or in the case where only single-grid data were used, the following three further feature selection methods were applied to determine the optimal number of features:

- FS1. A method based on the correlation coefficient between each feature and the output variable.
- FS2. A method based on the mutual information between each feature and the output variable.
- FS3. A method based on feature importance calculated by LightGBM.

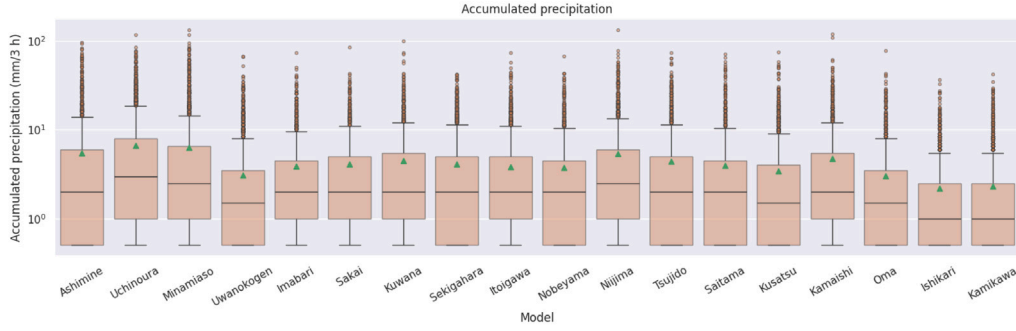
While many feature selection methods exist, this study adopted representative approaches (Chandrashekar and Sahin, 2014; Ross, 2014). The correlation coefficient-based method calculates the correlation coefficient between each feature and the output variable, and selects features in descending order of the correlation coefficient. The mutual information-based method uses mutual information between each feature and the output variable, where mutual information measures the degree of dependence between two variables (Ross, 2014). The LightGBM feature importance-based method calculates the contribution of each feature to reducing the loss function during tree splitting in LightGBM. Additionally, an analysis of the number of selected features was conducted. Variables such as prediction time, month, and hour were included in the final selected features regardless of the feature selection results.

2.4. Parameter tuning

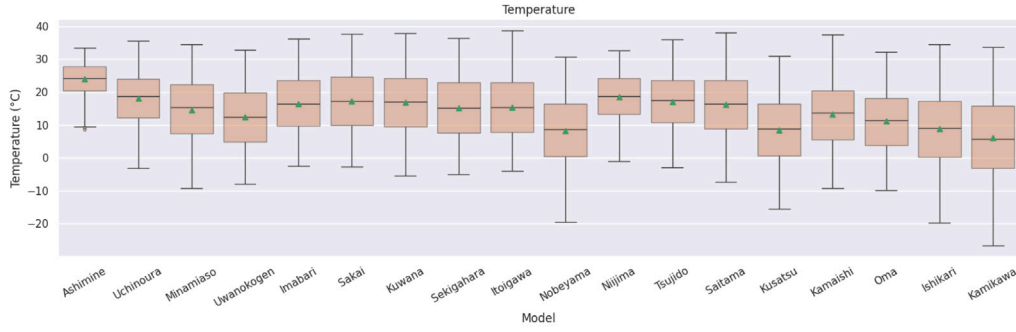
Parameter tuning was conducted using Optuna (Akiba et al., 2019), a parameter optimization library based on Bayesian optimization algorithms. Specifically, this study employed LightGBMTuner (OPTUNA, 2024) to compute optimal values for each parameter iteratively. The tuning process was performed using a single representative location, and the tuned parameters were applied to other locations. The chosen representative location was Sekigahara, situated near the center of all target locations. Since LightGBM cannot handle multiple output variables, tuning was conducted separately for each output variable.

2.5. Neural network and CNN

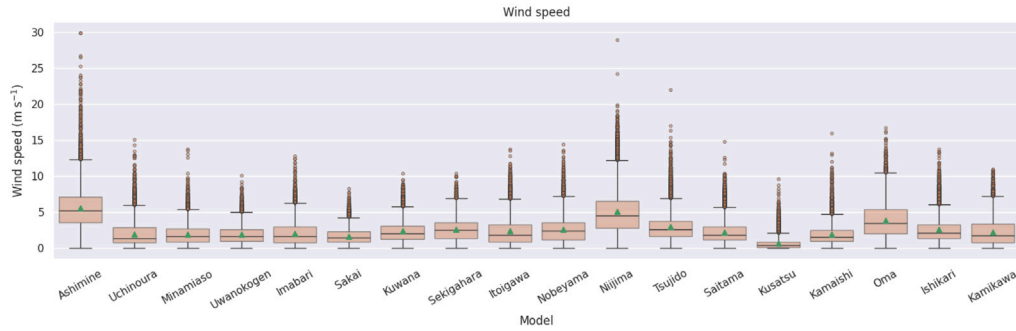
To provide comparisons with neural-network-based approaches used in previous studies, we also implemented a feedforward neural network



(a) Precipitation



(b) Temperature



(c) Wind Speed

Fig. 2. Box plots of the distributions of each meteorological variable at each location. For precipitation, only nonzero values are shown (values > 0 mm) to avoid the dominance of zero precipitation and to provide an informative visualization; the precipitation axis is displayed on a logarithmic scale. The green triangle markers indicate the mean values.

and a CNN model. The structure of the neural network, implemented using TensorFlow (TensorFlow, 2024) with keras.layers, is as follows:

- Dense(1000, activation='relu')
- Dense(500, activation='relu')
- Dense(100, activation='relu')
- Dense(3)

The input dimension corresponds to the number of features after FSO selection, and the output dimension (3) corresponds to precipitation, temperature, and wind speed.

The model was trained with the Adam optimizer (learning rate 0.001) (Kingma, 2014), MSE as the loss, and Root Mean Squared Error

(RMSE) :

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (2)$$

as a metric. Training used early stopping with a patience of 10 epochs, based on validation RMSE. We set a maximum of 1000 epochs with a batch size of 64; in practice, training typically converged within about 25–50 epochs due to early stopping.

CNN models have many possible variations, including the number of layers, kernel sizes, numbers of channels, the use of skip connections, and the choice of optimization and regularization methods. A comprehensive comparison of these architectures is beyond the scope of the present paper. Accordingly, we limited the comparison to a CNN model based on prior work that used CNNs for post-processing MSM data.

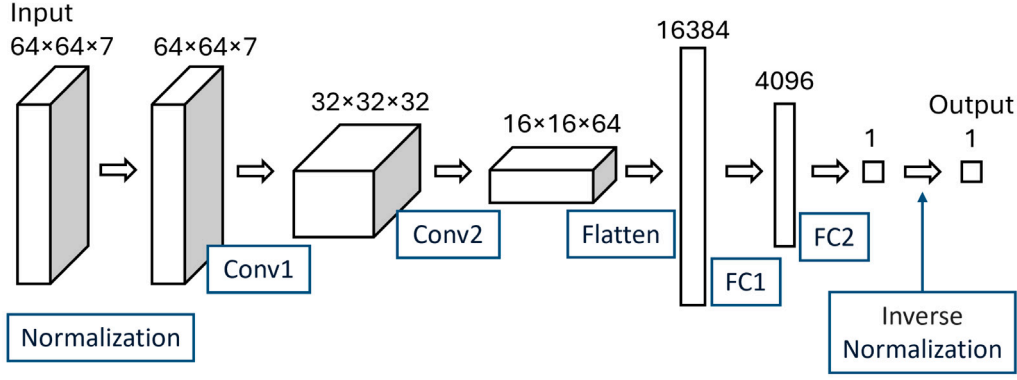


Fig. 3. Architecture of the CNN.

For the CNN baseline, we followed the experimental setting of Kudo (2022) to reproduce a representative prior CNN-based approach under comparable data conditions. Accordingly, the CNN was applied only to temperature prediction, as in the original study.

Our CNN uses a single forecast-time spatial field without a lookback window. The input is a $64 \times 64 \times 7$ tensor centered at each of the 18 locations. The seven channels consist of four surface-level fields (surface temperature, mean sea-level pressure, and surface wind components U and V) and temperature at three pressure levels (975, 925, and 850 hPa). Because the pressure-level grid is coarser than the surface grid, the pressure-level temperature fields are first cropped as 32×32 patches on the pressure-level grid and then upsampled to 64×64 to match the surface fields. The architecture of the CNN is shown in Fig. 3 and summarized in Table 4.

Normalization for CNN was performed as follows:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (3)$$

where x' represents the normalized value, x is the original value, and $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the MSM input data. These minimum and maximum values were calculated for each prediction time (data sample) rather than across the entire training dataset. For temperature, $x_{\min}$ was adjusted to $x_{\min} - 3$ and $x_{\max}$ to $x_{\max} + 3$. An inverse normalization step was applied after the output layer, defined as:

$$y' = y \cdot (x_{\max} - x_{\min}) + x_{\min} \quad (4)$$

where y is the output from the output layer, and y' is the value after inverse normalization. These normalization and inverse-normalization procedures were introduced to reproduce the prior study as closely as possible, and should thus be understood as part of the baseline implementation rather than as independently optimized design choices in this study.

The learning settings for CNN, including the optimizer, loss function, and batch size, were the same as for NN. We set a maximum of 1000 epochs; in practice, training typically converged within about 15–20 epochs due to early stopping.

2.6. Evaluation metrics

In addition to RMSE, we used Mean Error (ME),

$$ME = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i),$$

to assess model bias, where N is the number of data points, $\hat{y}_i$ is the predicted value, and y_i is the observed value.

In addition, the following metrics were used to further assess model performance for specific variables.

Table 4

Summary of processing at each CNN layer. Each layer's implementation is described using TensorFlow (TensorFlow, 2024) keras.layers.

Layer	Function
Conv1	ZeroPadding2D(padding=(2, 2))
	Conv2D(32, kernel_size=5, strides=1, padding='valid')
	MaxPooling2D(pool_size=2, strides=2)
	BatchNormalization()
	Activation('relu')
Conv2	ZeroPadding2D(padding=(2, 2))
	Conv2D(64, kernel_size=5, strides=1, padding='valid')
	MaxPooling2D(pool_size=2, strides=2)
	BatchNormalization()
	Activation('relu')
FC1	Dense(4096)
	BatchNormalization()
	Activation('relu')
FC2	Dense(1)

2.6.1. Statistical tests for temperature and wind speed

Statistical comparisons were conducted using paired absolute errors. Let

$$e_i^{\text{main}} = |\hat{y}_i^{\text{main}} - y_i|, \quad e_i^{\text{base}} = |\hat{y}_i^{\text{base}} - y_i|$$

denote the absolute errors of the main and baseline models for sample i . Then the mean absolute errors (MAEs) are defined as

$$\text{MAE}_{\text{main}} = \frac{1}{N} \sum_{i=1}^N e_i^{\text{main}}, \quad \text{MAE}_{\text{base}} = \frac{1}{N} \sum_{i=1}^N e_i^{\text{base}},$$

and the paired differences are defined by

$$d_i = e_i^{\text{main}} - e_i^{\text{base}}.$$

Based on these paired differences, we conducted a paired t -test and a Wilcoxon signed-rank test for temperature and wind speed. Because of the very large sample size, these tests are interpreted only as supplementary information, and practical significance is assessed mainly from the error metrics themselves.

2.6.2. Event-based metrics for precipitation

For precipitation, forecast skill was additionally evaluated using event-based metrics. For each precipitation threshold τ , forecasts and observations were converted into binary events. Let $H(\tau)$, $M(\tau)$, and $F(\tau)$ denote the numbers of hits, misses, and false alarms, respectively,

at threshold τ . Then, the event-based metrics were defined as

$$\begin{aligned} \text{TS}(\tau) &= \frac{H(\tau)}{H(\tau) + M(\tau) + F(\tau)}, \\ \text{POD}(\tau) &= \frac{H(\tau)}{H(\tau) + M(\tau)}, \\ \text{FAR}(\tau) &= \frac{F(\tau)}{H(\tau) + F(\tau)}, \\ \text{Bias}(\tau) &= \frac{H(\tau) + F(\tau)}{H(\tau) + M(\tau)}. \end{aligned}$$

2.7. Weighted tweedie model for precipitation

Precipitation is a nonnegative variable with a highly skewed distribution and many zero or near-zero values. Under such conditions, standard regression training based on mean squared error tends to be dominated by the large number of weak-rainfall or no-rainfall samples, and may therefore lead to predictions biased toward intermediate values, without sufficiently improving forecast accuracy for moderate to heavy precipitation events. To address these characteristics, in this study we examined a model that uses a Tweedie loss function with sample weight coefficients in LightGBM training for precipitation prediction (hereafter referred to as the weighted Tweedie model) (Jørgensen, 1997; Ke et al., 2017; Dunn and Smyth, 2005). In preliminary experiments for a few sites, the weighted Tweedie model showed more improvements than the variants using only weighting or only the Tweedie loss, so we focus on this variant here.

Let $y_i \geq 0$ denote the observed precipitation for sample i , and let $\mu_i > 0$ denote the predicted mean precipitation. The Tweedie distribution belongs to the exponential dispersion family and satisfies (Jørgensen, 1997; Dunn and Smyth, 2005)

$$\text{Var}(Y_i) = \phi \mu_i^p,$$

where $\phi > 0$ is the dispersion parameter and p is the Tweedie variance power. In this study, the exponent p was treated as a tunable hyperparameter.

The sample weights were defined as a monotone nondecreasing piecewise linear function of the observed precipitation. Let the knot positions be

$$x_0 = 0, \quad x_1 = 5, \quad x_2 = 10, \quad x_3 = 15, \quad x_4 = 20 \text{ mm},$$

and the corresponding weight values be

$$w_0, w_1, w_2, w_3, w_4.$$

A monotonicity constraint,

$$1 = w_0 \leq w_1 \leq w_2 \leq w_3 \leq w_4,$$

was imposed. Then, the sample weight function $w(y)$ was defined by linear interpolation between adjacent knots as follows:

$$w(y) = \begin{cases} w_0 + \frac{w_1 - w_0}{x_1 - x_0}(y - x_0), & x_0 \leq y \leq x_1, \\ w_1 + \frac{w_2 - w_1}{x_2 - x_1}(y - x_1), & x_1 < y \leq x_2, \\ w_2 + \frac{w_3 - w_2}{x_3 - x_2}(y - x_2), & x_2 < y \leq x_3, \\ w_3 + \frac{w_4 - w_3}{x_4 - x_3}(y - x_3), & x_3 < y. \end{cases}$$

In the implementation, the resulting weights were clipped to the range

$$1 \leq w(y) \leq w_{\max},$$

where $w_{\max} = 10$. Thus, samples with higher precipitation intensity contributed more strongly to training, while the monotonicity constraint maintained the stability and interpretability of the weighting scheme.

Using these weights, the precipitation model was trained by minimizing the weighted Tweedie loss

$$\mathcal{L}_{\text{train}} = \sum_{i=1}^N w(y_i) \ell_{\text{TW}}(y_i, \mu_i),$$

where $\ell_{\text{TW}}(y_i, \mu_i)$ denotes the Tweedie loss for sample i . In this study, we used the Tweedie objective implemented in LightGBM (Ke et al., 2017; LightGBM Contributors, 2026). Except for the precipitation loss function and sample-weighting scheme described below, the input data configuration and the tree-related LightGBM settings were the same as those of the “around all tune” model.

Although the model was trained using the weighted Tweedie loss, the hyperparameters were selected in Optuna based on an event-based evaluation score, because the purpose of weighting was to improve performance at relevant precipitation thresholds. The tuned hyperparameters were the Tweedie variance power p and the parameters determining the shape of the weight function, and the tuning was carried out separately for each observation site because the behavior of the weighted Tweedie model showed strong site dependence. A monotone nondecreasing constraint was imposed on the weight function so that the weight would not decrease as precipitation intensity increased.

Using the event-based metrics defined in Section 2.6.2, the TS values across thresholds were aggregated using a geometric mean:

$$S_{\text{geo}} = \exp\left(\frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \log(\max(\text{TS}(\tau), \epsilon))\right).$$

Furthermore, to suppress overprediction or underprediction of event frequency, a penalty was imposed on the forecast Bias at selected thresholds. Using this, the total score employed in Optuna was defined as

$$S_{\text{total}} = S_{\text{geo}} - \sum_{\tau \in \mathcal{T}_{\text{bias}}} \gamma_{\tau} \left| \log(\max(\text{Bias}(\tau), b_{\min})) \right|.$$

Here, $\gamma_{\tau} \geq 0$ denotes the penalty coefficient, $\mathcal{T}_{\text{bias}}$ denotes the set of thresholds to which the bias penalty is applied, and $b_{\min}$ is a small positive constant for numerical stability. This objective function was intended to encourage models with balanced detection skill across multiple thresholds while reducing substantial overprediction or underprediction of event frequency.

In the present implementation, the thresholds used for optimization were set to

$$\mathcal{T} = \{1, 2, \dots, 20\} \text{ mm},$$

and equal weights were used for all thresholds in the geometric mean. In addition, the bias penalty was applied at 1, 5, 10, and 15 mm with $\gamma_{\tau} = 0.05$ for all of these thresholds, and $\epsilon = 10^{-6}$ and $b_{\min} = 10^{-6}$ were set for numerical stability. The resulting model is referred to as the weighted Tweedie model in the following sections.

3. Results and discussion

3.1. Model comparison

3.1.1. RMSE and ME

First, Tables 5 and 6 present the RMSE and ME of each model, as well as for MSM and MSMG, on the validation and test datasets.

For RMSE, both validation and test datasets show that the models using LightGBM outperform MSM and MSMG. Similarly, the neural network and CNN models perform worse compared to those using LightGBM. Moreover, models incorporating data from surrounding grids (around) exhibit slightly better performance than those using only the nearest grid (1grid). This indicates that leveraging features from surrounding grids and reducing them through feature selection allows for the inclusion of informative data that improves accuracy. The correlation analysis-based feature selection (FS0) of surrounding grid data yielded the best results in most cases. Among the three feature selection methods (FS1 - FS3) described in Section 2.3, no performance improvement was observed. For the number of features in FS1 to FS3, we used the values that performed best for each feature selection method and each output variable in Section 3.2.

Table 5

RMSE and ME for validation data. “1grid” represents models using data from only one grid, “around” indicates the use of surrounding grid data, “nn” denotes neural network, “all” refers to data before three feature selection methods (FS1 - FS3), “pcc” indicates the correlation coefficient-based method (FS1), “mi” denotes the mutual information method (FS2), “gbm” represents the feature importance method by LightGBM (FS3), and “tune” refers to parameter tuning. Averages are calculated across prediction times and locations.

	RMSE			ME		
	Precip.	Temp.	Wind	Precip.	Temp.	Wind
MSM	2.948	1.836	2.096	0.048	-0.405	0.753
1grid nn	2.555	1.472	1.167	-0.075	0.130	-0.013
1grid all	2.510	1.364	1.070	-0.007	0.084	-0.018
1grid pcc	2.526	1.367	1.070	-0.003	0.085	-0.018
1grid mi	2.535	1.420	1.070	-0.001	0.096	-0.016
1grid gbm	2.518	1.363	1.073	-0.006	0.085	-0.018
around all	2.501	1.332	1.050	0.001	0.081	-0.025
around pcc	2.496	1.344	1.053	0.006	0.082	-0.026
around mi	2.505	1.342	1.051	0.004	0.083	-0.026
around gbm	2.502	1.333	1.051	-0.002	0.080	-0.027
around all tune	2.489	1.326	1.046	0.009	0.074	-0.023
CNN		1.487			0.086	

Table 6

RMSE and ME for test data.

	RMSE			ME		
	Precip.	Temp.	Wind	Precip.	Temp.	Wind
MSM	2.759	1.811	1.999	0.085	-0.340	0.673
MSMG	2.726	1.394	1.237	0.107	0.014	0.159
1grid nn	2.439	1.485	1.175	-0.064	0.143	-0.060
1grid all	2.387	1.368	1.075	0.020	0.110	-0.058
1grid pcc	2.390	1.369	1.076	0.016	0.109	-0.058
1grid mi	2.409	1.428	1.075	0.028	0.117	-0.057
1grid gbm	2.392	1.368	1.078	0.017	0.111	-0.058
around all	2.358	1.341	1.053	0.031	0.111	-0.062
around pcc	2.374	1.349	1.056	0.030	0.124	-0.062
around mi	2.381	1.347	1.055	0.033	0.119	-0.061
around gbm	2.361	1.341	1.052	0.028	0.112	-0.063
around all tune	2.344	1.335	1.052	0.033	0.106	-0.059
CNN		1.501			0.179	

For ME, models using LightGBM tended to reduce bias in both validation and test datasets. However, for precipitation in the validation dataset, MSM achieved the best ME, while for temperature in the test dataset, MSMG performed the best.

For subsequent predictions in this study, we used the “around all tune” model, which showed the best overall performance; here, “around” indicates the use of surrounding-grid predictors, “all” indicates that all selected predictors were retained without further reduction, and “tune” indicates hyperparameter optimization.

Fig. 4 shows the prediction results for test data compared with MSM, MSMG, and observed values at selected locations and prediction times. In the case of 3-hour precipitation at Kamikawa and 18-hour wind speed at Itoigawa, post-processing models corrected the apparent large bias. On the other hand, for 18-hour temperature at Itoigawa, partial corrections by the post-processing model were observed.

3.1.2. Statistical tests and event-based metrics

For temperature and wind speed, Table 7 summarizes the paired t -test and Wilcoxon signed-rank test results comparing the main model (around all tune) with the baseline models (MSM and MSMG), based on the metrics defined in Section 2.6.1. The table reports MAE_{main} , MAE_{base} , the standard deviation of the paired differences, and the corresponding one-sided p -values. Because of the very large sample size, these test results should be interpreted only as supplementary information, and practical significance should be assessed mainly from the error metrics themselves.

For precipitation, Table 8 summarizes the event-based metrics defined in Section 2.6.2—TS, POD, FAR, and Bias—for four precipitation

Table 7

Paired t -test and Wilcoxon signed-rank test results comparing the LightGBM “around all tune” model (main model) with the baseline models for temperature and wind speed. The table reports the mean absolute errors (MAEs) of the main and baseline models, the standard deviation of the paired differences in absolute error (SD_{diff}), and the corresponding one-sided p -values. All tests were performed over $N = 682,305$ paired samples.

Variable	Baseline	MAE_{main}	MAE_{base}	SD_{diff}	p_t	p_W
Temperature	MSM	0.9888	1.3964	1.0840	0.000	0.000
Temperature	MSMG	0.9888	1.0275	0.7675	0.000	0.000
Wind Speed	MSM	0.7549	1.4186	1.3530	0.000	0.000
Wind Speed	MSMG	0.7549	0.9035	0.6980	0.000	0.000

Table 8

Event-based verification metrics for precipitation, pooled over all locations and forecast lead times. Higher TS and POD indicate better detection skill, lower FAR indicates fewer false alarms, and Bias values closer to 1 indicate better frequency matching. Here, “around all tune” denotes the original LightGBM-based post-processing model.

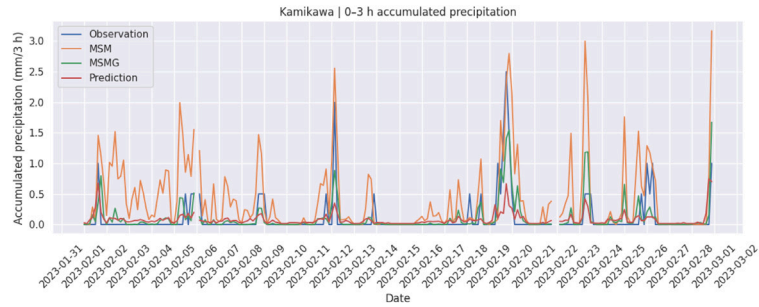
Threshold (mm)	Model	TS	POD	FAR	Bias
1.0	MSM	0.4311	0.6373	0.4287	1.1156
1.0	MSMG	0.4555	0.6558	0.4014	1.0956
1.0	around all tune	0.4293	0.6931	0.4699	1.3075
1.0	weighted Tweedie	0.4435	0.7005	0.4527	1.2800
5.0	MSM	0.3226	0.5061	0.5293	1.0753
5.0	MSMG	0.3344	0.5014	0.4991	1.0009
5.0	around all tune	0.3059	0.3943	0.4228	0.6831
5.0	weighted Tweedie	0.3316	0.5208	0.5228	1.0912
10.0	MSM	0.2450	0.4084	0.6201	1.0749
10.0	MSMG	0.2485	0.4122	0.6152	1.0713
10.0	around all tune	0.1966	0.2356	0.4571	0.4340
10.0	weighted Tweedie	0.2520	0.3979	0.5927	0.9771
15.0	MSM	0.1902	0.3189	0.6796	0.9956
15.0	MSMG	0.2096	0.3551	0.6616	1.0492
15.0	around all tune	0.1251	0.1420	0.4877	0.2772
15.0	weighted Tweedie	0.1883	0.2719	0.6201	0.7158

thresholds. Here, “around all tune” denotes our original LightGBM-based post-processing model, and “weighted Tweedie” denotes the weighted Tweedie model.

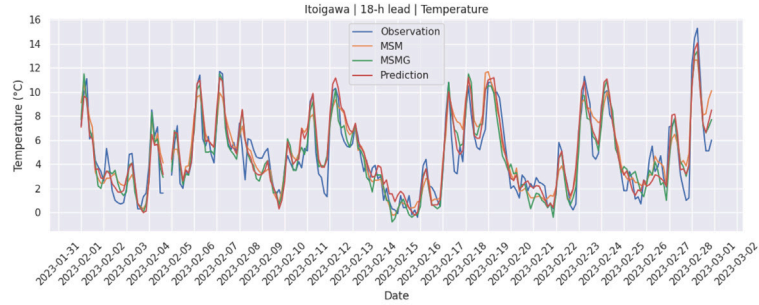
Overall, MSMG showed favorable performance among the compared models. The “around all tune” model achieved the highest POD at the 1.0 mm threshold, indicating high sensitivity to light precipitation events. However, its FAR and Bias were also the largest, suggesting that it tended to overpredict precipitation occurrence. At thresholds of 5.0 mm and above, its TS and POD became lower than those of MSM and MSMG, and its Bias fell well below 1 at 10.0 mm and 15.0 mm, indicating underdetection of heavier precipitation events. Compared with this behavior, the weighted Tweedie model improved TS and Bias at 10.0 mm and increased POD at 5.0 mm. Although its overall performance was still slightly below that of MSMG, it showed a better balance across the precipitation thresholds.

3.2. Comparison of feature selection methods

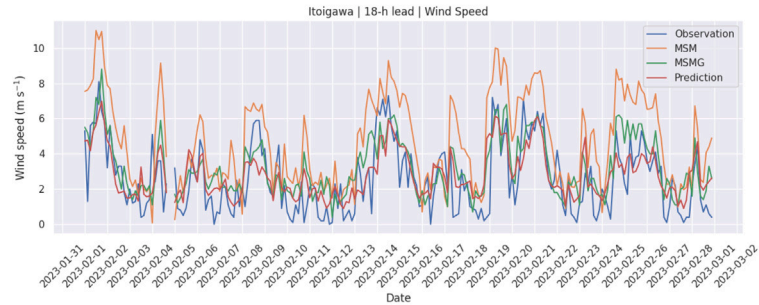
Here, we conducted experiments at a single location in Sekigahara, similar to parameter tuning, to evaluate the influence of the number of selected features in the three feature selection methods (FS1 - FS3) proposed in Section 2.3. For both cases where only the data of the grid closest to the prediction point (1grid) and the data including surrounding grids (around) were used, the three feature selection methods were applied, and models were constructed by incrementally increasing the number of selected features k by 10. Fig. 5 summarizes the RMSE on the validation data for each feature count k . The results indicate that the accuracy tends to improve as the number of features increases, but stabilizes when the number of features exceeds a certain threshold. This suggests that further feature selection after correlation analysis does not enhance accuracy.



(a) Accumulated precipitation for the 0–3 h forecast interval at Kamikawa



(b) 18-h lead temperature at Itoigawa



(c) 18-h lead wind speed at Itoigawa

Fig. 4. Examples of prediction results for the test data in February, compared with MSM, MSMG, and observed values. Panel (a) shows accumulated precipitation for the 0–3 h forecast interval, while panels (b) and (c) show temperature and wind speed at the 18-h forecast lead time, respectively.

3.3. Results by location and prediction time

3.3.1. Location-based results

Fig. 6 summarizes the RMSE on test data for each observation location. Our post-processing model consistently outperformed MSM and showed comparable or slightly better performance than MSMG across most sites. For precipitation, relatively large RMSE values were observed in the southwestern part of Japan, including Ashimine and Uchinoura. This is likely associated with the frequent occurrence of typhoons and strong convective rainfall events in these regions. In contrast, the lowest precipitation RMSE values were found in inland and mountainous areas such as Kusatsu and Nobeyama.

For wind speed, extremely high RMSE values in MSM were observed in island and coastal areas such as Nijima and Oma. Our model significantly reduced these errors, achieving RMSE values comparable to those in other regions. Overall, the regional pattern of RMSE reduction indicates that the post-processing model effectively mitigates the biases and noise that are spatially correlated with topography and surrounding meteorological conditions.

As shown in Table 9, the selected *around_all_tune* model achieved the lowest RMSE across all terrain types and variables. The improvement was most pronounced in mountainous and island areas, where spatial

Table 9

Average RMSE by terrain category and model. The three terrain categories are defined as follows: **mountain** (Kusatsu, Kamikawa, Nobeyama, Uwanokogen, Minamiaso), **island** (Ashimine, Nijima), and **coastal/plain** (Ishikari, Oma, Imabari, Kamaishi, Tsujido, Itoigawa, Uchinoura, Sekigahara, Saitama, Kuwana, Sakai).

Region	around_all_tune			MSM			MSMG		
	Precip.	Temp.	Wind	Precip.	Temp.	Wind	Precip.	Temp.	Wind
coastal/plain	1.982	1.343	1.002	2.454	1.793	1.864	2.400	1.363	1.211
island	2.951	1.003	1.576	3.492	1.654	2.550	3.349	1.106	1.647
mountain	2.369	1.398	0.822	2.630	1.804	1.337	2.605	1.508	1.038

heterogeneity in meteorological conditions is large. This demonstrates that incorporating spatial context from surrounding grids improves model robustness under complex terrain and coastal influences.

3.3.2. Prediction-time-based results

Fig. 7 summarizes the RMSE on test data for each prediction time. In terms of RMSE, the selected *around_all_tune* model outperformed both MSM and MSMG at almost all forecast lead times. For all methods, RMSE increased as prediction time became longer, which is a common

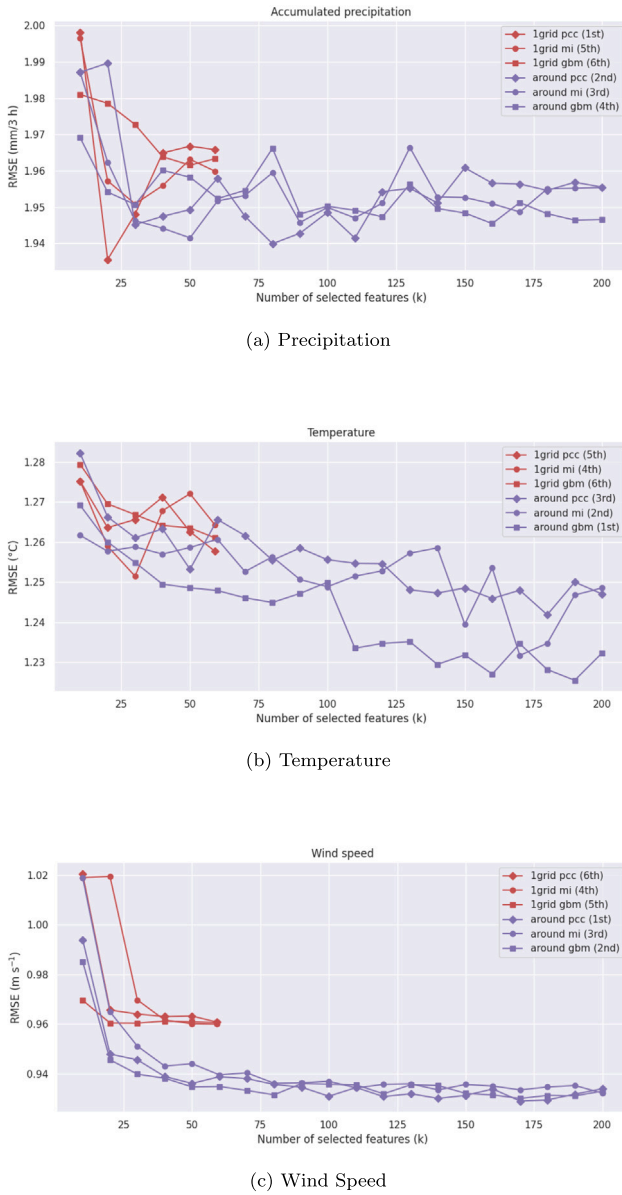


Fig. 5. Changes in accuracy based on the number of selected features for the feature selection methods on validation data in Sekigahara. Results are averaged over prediction times.

characteristic of numerical weather prediction (NWP) models, where forecast accuracy declines with lead time. A similar trend was observed in our post-processing results, indicating that the residual errors of the base NWP propagate into the corrected predictions as forecast uncertainty accumulates. Nevertheless, the selected `around_all_tune` model maintained lower RMSE than MSM and, in many cases, MSMG across forecast horizons, confirming its stability over time.

3.3.3. Scatter plots against observations

Fig. 8 compares observations with the model outputs at Sekigahara during the test period. For readability, all lead times are pooled and a random subset of 150 samples is plotted.

3.4. Site-dependent behavior of the weighted tweedie model

The effect of the weighted Tweedie model was strongly site dependent. To illustrate this point, we compare two representative examples:

Table 10

Site-wise precipitation verification metrics for two representative cases. Higher TS and POD indicate better detection skill, while Bias values closer to 1 indicate better frequency matching.

Site	Model	Threshold (mm)	TS	POD	Bias
Kamikawa	MSMG	1.0	0.3802	0.5501	0.9933
	around all tune	1.0	0.3817	0.5311	0.9367
	weighted Tweedie	1.0	0.3863	0.5319	0.9276
	MSMG	5.0	0.2447	0.3897	0.9464
	around all tune	5.0	0.1541	0.1852	0.3492
	weighted Tweedie	5.0	0.1264	0.1488	0.3057
	MSMG	10.0	0.1278	0.2802	0.9973
	around all tune	10.0	0.0267	0.0275	0.0687
	weighted Tweedie	10.0	0.0471	0.0522	0.0962
	MSMG	15.0	0.0667	0.1703	0.9725
	around all tune	15.0	0.0000	0.0000	0.0000
	weighted Tweedie	15.0	0.0055	0.0055	0.0055
Saitama	MSMG	1.0	0.4391	0.6383	1.0919
	around all tune	1.0	0.4150	0.6911	1.3563
	weighted Tweedie	1.0	0.4330	0.7160	1.3696
	MSMG	5.0	0.3549	0.4994	0.9065
	around all tune	5.0	0.2308	0.2521	0.3444
	weighted Tweedie	5.0	0.3429	0.4959	0.9420
	MSMG	10.0	0.1935	0.3231	0.9923
	around all tune	10.0	0.1493	0.1538	0.1846
	weighted Tweedie	10.0	0.3395	0.4962	0.9577
	MSMG	15.0	0.2048	0.3675	1.1624
	around all tune	15.0	0.0678	0.0684	0.0769
	weighted Tweedie	15.0	0.2600	0.3333	0.6154

Kamikawa, where the improvement was limited, and Saitama at the 18-h lead time, where the weighted Tweedie model showed clearer improvement in the time-series comparison.

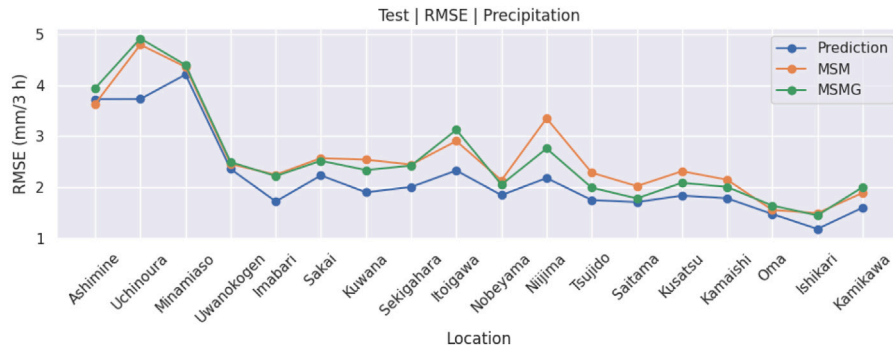
Fig. 9(a) shows the 0–3 h accumulated precipitation at Kamikawa. In this case, the weighted Tweedie model remained close to the original “around all tune” model, and the difference between the two predictions was small for many events. This behavior is consistent with the site-wise aggregated verification results summarized in Table 10. At Kamikawa, the weighted Tweedie model produced only a slight change at the 1.0 mm threshold and did not improve TS at 5.0 mm, although TS and POD were slightly improved at 10.0 and 15.0 mm. Overall, however, the weighted Tweedie model still remained clearly below MSMG for this site, indicating that the modification was not sufficient to overcome the difficulty of precipitation forecasting at this site.

By contrast, Fig. 9(b) shows a case at Saitama for the 15–18 h accumulated precipitation, where the weighted Tweedie model more clearly modified the predicted peaks and, for several events, tracked the observed precipitation more closely than “around all tune”. This behavior is also reflected in the site-wise aggregated results in Table 10. At Saitama, the weighted Tweedie model improved TS relative to “around all tune” at all four thresholds and substantially increased POD at 5.0, 10.0, and 15.0 mm, while also moving Bias much closer to 1 than the original model at 5.0 and 10.0 mm. At 15.0 mm, the weighted Tweedie model still showed the best TS among the compared models, although its Bias remained below 1. These results suggest that the weighted Tweedie model can improve event-oriented behavior for some sites, especially when the original model underdetects moderate to heavy precipitation.

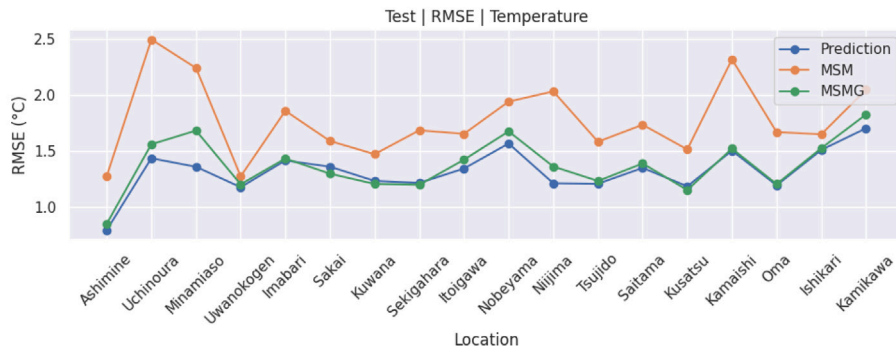
Taken together, these examples indicate that the weighted Tweedie model is useful mainly as a site-dependent adjustment rather than as a uniformly superior replacement for the original LightGBM model. This observation is consistent with the need for site-wise hyperparameter tuning discussed in Section 2.7.

4. Conclusion

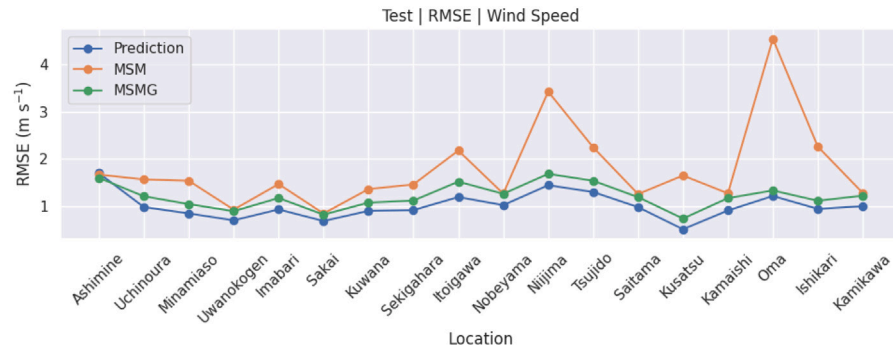
In this study, we developed machine learning-based post-processing models for precipitation, temperature, and wind speed using observation data and MSM data across 18 locations in Japan, including plains,



(a) Precipitation



(b) Temperature



(c) Wind Speed

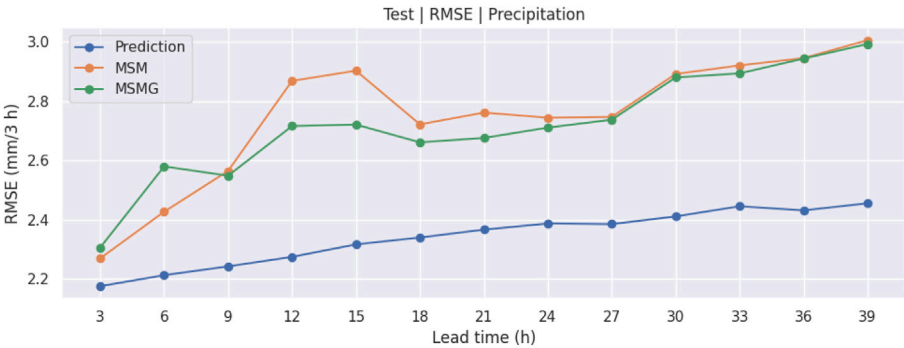
Fig. 6. RMSE by location on test data. Results are averaged over prediction times.

mountainous areas, and islands. The main findings are summarized as follows:

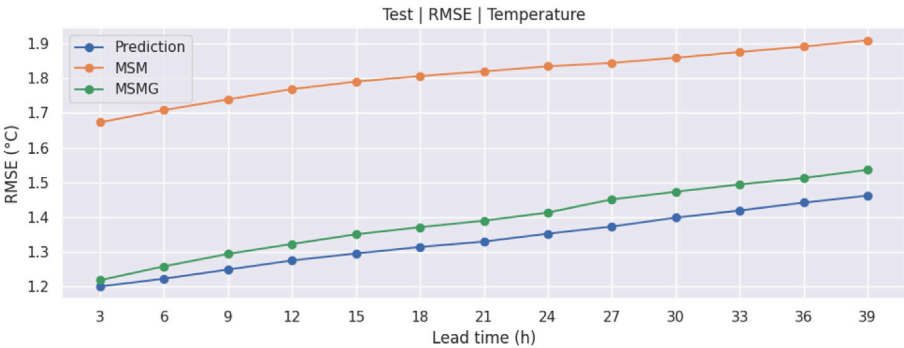
1. The use of surrounding grid data together with correlation-based feature selection improved prediction accuracy compared with using only the single nearest grid. This result supports the effectiveness of incorporating spatial information from surrounding grids while controlling feature dimensionality through feature selection. In contrast, further feature selection beyond this correlation-based reduction did not provide additional improvement in our experiments.
2. In our experimental setting, the LightGBM-based models achieved lower RMSE than the tested neural-network base-

lines, including the reproduced CNN baseline, and also generally achieved lower RMSE than the JMA guidance product MSMG. However, because the neural-network comparison was limited to specific baseline implementations, the present results do not exclude the possibility that stronger deep-learning architectures could perform better.

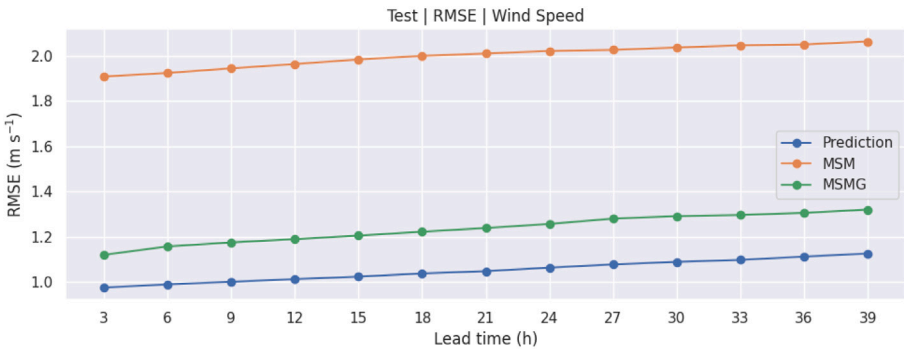
3. For precipitation, the results showed that optimizing only standard regression-oriented objectives is not necessarily sufficient, especially when the forecasting goal is to improve the prediction of moderate and heavy rainfall events. To address this issue, we introduced Tweedie-based loss functions and event-weighted training strategies. These approaches improved event-oriented behavior, including better detection of stronger rainfall events



(a) Precipitation



(b) Temperature



(c) Wind Speed

Fig. 7. RMSE by prediction time on test data. Results are averaged over locations.

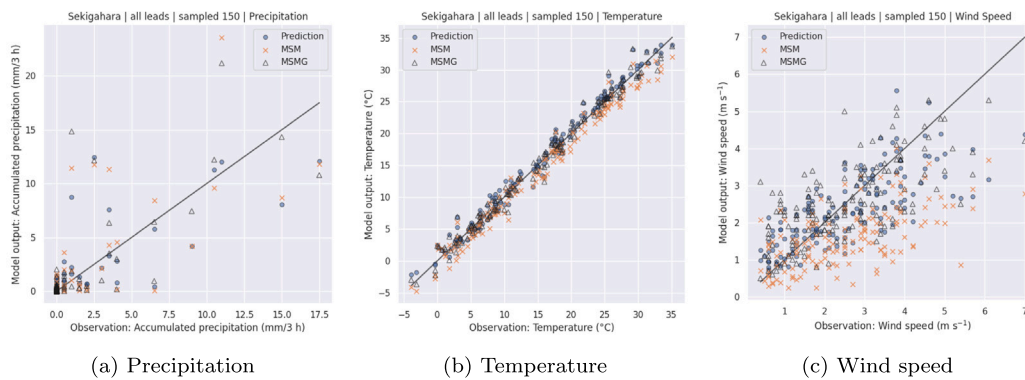


Fig. 8. Scatter plots at Sekigahara on the test period (all lead times pooled). Each panel shows observations versus model outputs for the same randomly sampled points.

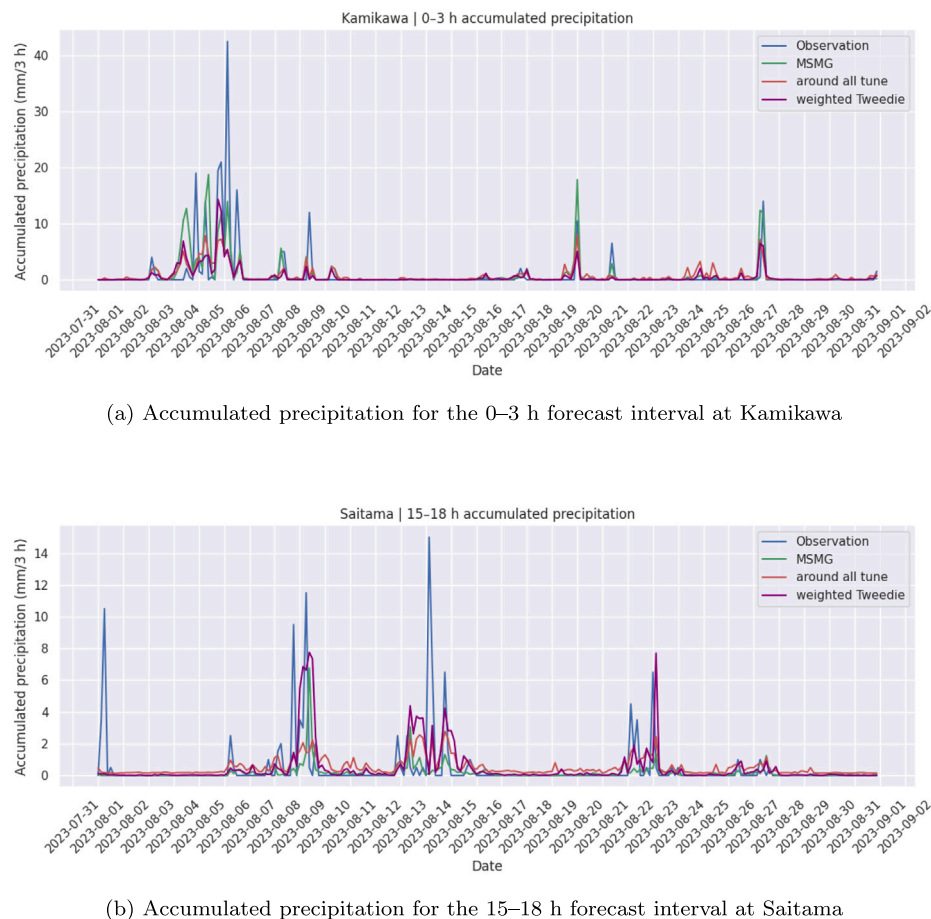


Fig. 9. Comparison of precipitation forecasts for two representative cases. Panel (a) shows Kamikawa, where the weighted Tweedie model remained close to the original “around all tune” model and the improvement was limited. Panel (b) shows Saitama, where the weighted Tweedie model showed improved event-oriented behavior for several precipitation events.

at some sites and thresholds. However, the gains were not uniform across locations and conditions, indicating that further improvement is still needed. This site dependence suggests that inter-station differences related to topography and site characteristics, such as latitude, longitude, and altitude, may need to be considered more explicitly in future work.

To complement the representative examples and pooled results shown in the main text, detailed verification results by site and forecast lead time are provided in the Supplementary Material. These include RMSE and ME of MSM, MSMG, and “around all tune” for precipitation,

temperature, and wind speed, as well as precipitation event-based metrics (TS, POD, FAR, and Bias), including those of the weighted Tweedie model.

CRediT authorship contribution statement

Kazuma Iwase: Writing – original draft, Methodology, Data curation, Conceptualization. **Tomoyuki Takenawa:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used ChatGPT to assist with code editing and language proofreading. All AI-assisted outputs were reviewed and revised by the authors as needed, and the authors take full responsibility for the content of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.atmosres.2026.109037>.

Data availability

The code used in this study is available at: <https://github.com/ttakenawa/PostMSM>. Due to data licensing restrictions, the repository provides a synthetic/derived dataset for demonstration, which may not exactly reproduce the paper results.

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